Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/17/2013
PI Name:	Stone, Lee S. Ph.D.		
Project Title:	Robust Human-System Interface Design for Spaceflight Induced Environments		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Leland.S.Stone@nasa.gov	**Fax:**	FY 650-604-0255
PI Organization Type:	NASA CENTER	**Phone:**	650-604-3240
Organization Name:	NASA Ames Research Center		
PI Address 1:	Human-Systems Integration Division		
PI Address 2:	Mail Stop 262-2		
PI Web Page:			
City:	Moffet Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	05/01/2009	**End Date:**	04/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kaiser, Mary (NASA Ames Research Center) Cowings, Patricia (NASA Ames Research Center) Toscano, Bill (NASA Ames Research Center) Sweet, Barbara (NASA Ames Research Center) Adelstein, Bernard (NASA Ames Research Center) Liston, Dorion (NASA Ames Research Center)		
Grant/Contract No.:	NCC 9-58-SA02002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The next generation of human-rated spaceflight vehicles will expose astronauts to altered gravito-inertial forces and elevated vibration levels during launch and landing that can affect their ability to perform their assigned tasks. Furthermore, current plans call for crew action in critical tasks, including the override of flight automation in off-nominal conditions, while the crew may be exposed to 4-times normal gravity or more combined with vibration levels potentially higher than those experienced during Apollo and Shuttle. The risk associated with these dynamic phases of flight becomes even more acute when critical tasks are performed by de-conditioned crew returning after a long-duration spaceflight. Altered gravitational environments experienced in spacecraft impact human manual and gaze control and high-frequency vibration is also a serious challenge to visuomotor performance and gaze control. Unfortunately, our current quantitative knowledge of these effects is too limited to develop and validate operationally relevant predictive models of crew performance of sufficient reliability to support interface design or requirement verification. In particular, the interactions between increased gravity and vibration are poorly understood. The goal of the proposed research is to measure, analyze, and model human performance in display navigation and flight control tasks during centrifugation and vibration (in isolation and combination) to simulate ascent/descent conditions of spaceflight. The proposed research will measure performance (and associated workload) in operationally relevant display navigation and manual control tasks under simulated dynamic phases of flight and will compare the effectiveness of a variety of potential display and interface configurations. These data will enable us to identify and model the human capabilities and limitations in order to provide validated quantitative design guidelines for displays and controls to support crew-cockpit design engineers and operations/training development. This research will enable the development of validated spacecraft interface design guidelines and human-system verification tools with the ultimate goal of reducing mission risks and enhancing safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will determine the impacts of transient and sustained acceleration on human sensorimotor performance. The benefit of this research for NASA is that it will fill a critical knowledge gap that is currently hampering the optimal design of astronaut and pilot displays and interfaces needed to maximize performance and safety and minimize human error. The benefit of this research for humanity above and beyond enabling safer and more productive space exploration is two-fold. First, the human performance databases (and models) generated will allow for better design of aircraft and other vehicles thereby ultimately facilitating the enhancement of safety for the traveling public by supporting improved cockpit/interface design and training. Second, the performance assessment technologies developed and validated in this project may have medical applications for enhancing diagnostic methods to identify sensorimotor deficits due to disease or injury.
Task Progress:	We have completed Specific Aims 1, 2, and 4 measuring the effects of elevated Gx-loads and vibration on human oculomotor, manual, and autonomic behavior to assess the impacts on vestibular, visual, and motor function and workload of induced environments representative of dynamic phases of flight. • We found that gaze stabilization (via the Vestibulo-Ocular Reflex) is not the limiting factor for sensorimotor control with Gx loads up to 3.8Gx and vibration levels of 0.5g at up to 16 Hz. We found that retinal blur is typically about 0.25 deg or less, indicating that the combined linear and translational Vestibulo-Ocular Reflex is nearly perfectly compensatory, even under challenging Gx-load and vibration conditions. • By comparing open-loop vs. closed-loop saccadic and reach performance, we found that: - 3.8-Gx loading significantly increases systematic errors for both saccades and manual reaches, compared to performance at 1 Gx. - Both saccadic and reaching accuracy and precision improved under closed-loop conditions as compared to open-loop conditions, indicating that oculomotor and manual control performance is visually limited under open-loop conditions (because of visual spatial uncertainty). - Vibration significantly decreases the precision of both saccades and reaches. - There is a significant correlation between the errors in eye and reach end points, indicating that at least some of the error is due to shared neural noise in visual localization for both oculomotor and manual control. • By comparing pursuit tracking performance in an upright (+1 Gz) vs. a supine (+1 Gx) posture, and under low (+1 Gx) vs. high (+3.8 Gx) G-loads, we found that visual motion processing is largely robust to altered G conditions with some exceptions: - The cardinal-oblique anisotropy (better performance for target motion along the cardinal directions than for those along the oblique direction) is insensitive to changes in the direction and magnitude of the G vector; - However, the horizontal-vertical asymmetry (typically higher direction gains for vertical than for horizontal target motion) is significantly flattened at 3.8 vs. 1 Gx, showing that this visual processing non-linearity is due to a different (possibly cortical) mechanism that is modulated by vestibular cues. • Elevated Gx-loads and vibration generate changes in Autonomic Nervous System (ANS) function, consistent with increased workload: - Both heart and breathing rates are significantly elevated during exposure to +3.8 Gx, as compared to +1.0 Gx. - We found a correlation between heart and breathing rate and response time showing that ANS effects are linked to task performance. - Heart and breathing are significant altered by vibration frequency, although dramatically differently for the two Gx levels. - This observed non-linear interaction between Gx-loading and vibration confirms why one must measure them in combination, as we have done in this project.
Bibliography Type:	Description: (Last Updated: 10/30/2019)

Articles in Peer-reviewed Journals	Liston DB, Stone LS. "Saccadic brightness decisions do not use a difference model." Journal of Vision. 2013 Jul 1;13(8):1. http://dx.doi.org/10.1167/13.8.1 ; PubMed PMID: 23818677 , Jul-2013
Articles in Peer-reviewed Journals	Godinez A, Liston DB, Ayzenberg R, Toscano WB, Cowings PA, Stone LS. "G-loading and vibration effects on heart and respiration rates." Aviat Space Environ Med. 2014 Sep;85(9):949-53. https://doi.org/10.3357/ASEM.4015.2014 ; PubMed PMID: 25197894 , Sep-2014
Articles in Peer-reviewed Journals	Liston DB, Adelstein BD, Stone LS. "Onset of positional vertigo during exposure to combined G loading and chest-to-spine vibration." Aviat Space Environ Med. 2014 Feb;85(2):183-6. PubMed PMID: 24597164 , Feb-2014
Articles in Peer-reviewed Journals	Liston DB, Stone LS. "Oculometric assessment of dynamic visual processing." J Vis. 2014 Dec 19;14(14):12. https://doi.org/10.1167/14.14.12 ; PubMed PMID: 25527150 , Dec-2014